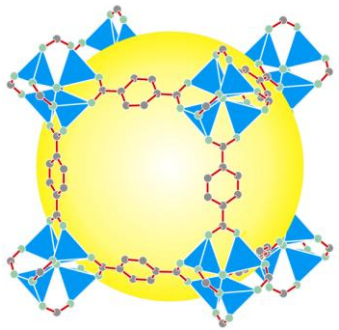


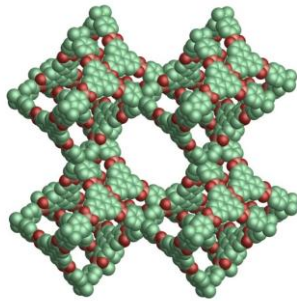
The Evolution of Metal-Organic Frameworks

Reticular chemistry
a. MOFs & Porous materials

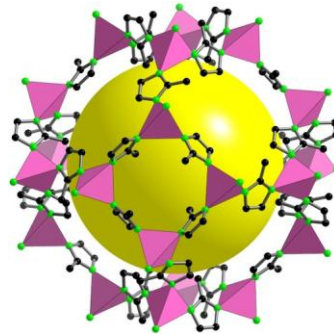
Molecules  Frameworks  Applications



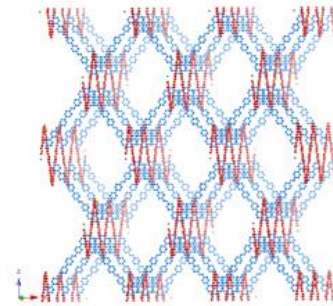
MOFs
[*Nature* 1995](#), [*JACS* 1998](#),
[*Nature* 1999](#)



COFs
[*Science* 2005](#), [*2007*](#)



ZIFs
[*PNAS* 2006](#)



Molecular Weaving
[*Science* 2016](#)

[Nobel Prize in Chemistry 2025](#)



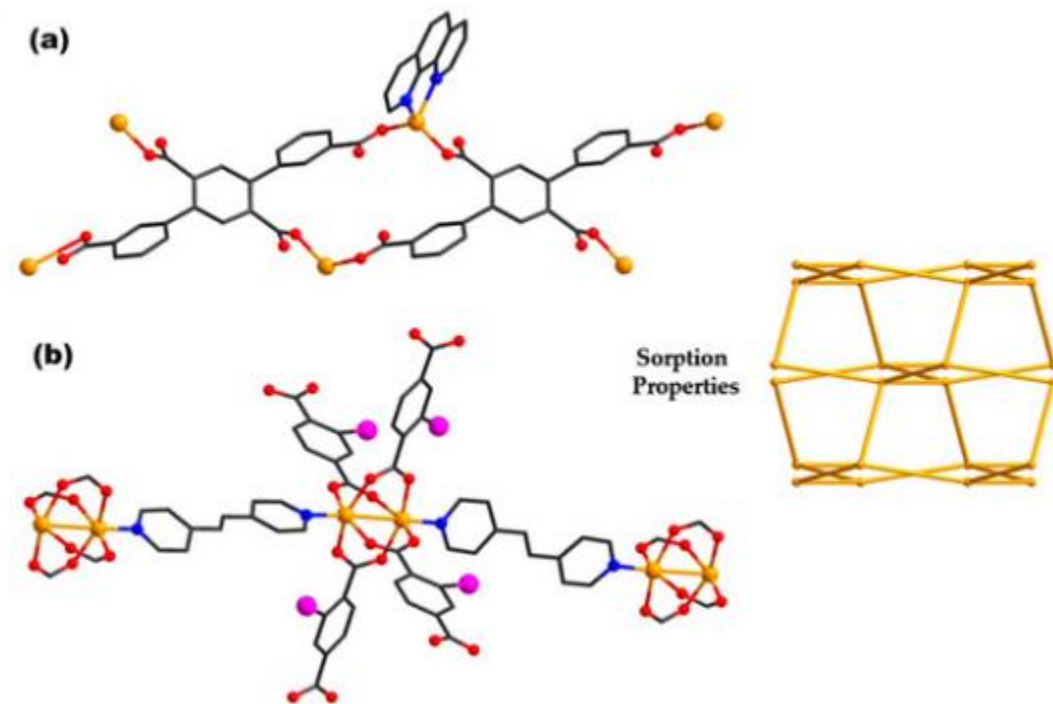
Prof. Omar M. Yaghi

1965: The Early Foundations

Zn-Coordination Polymers

Researchers at **DuPont** synthesized thermally stable coordination polymers using Zinc nodes.

While these materials showed the potential for repetitive bonding between metals and ligands, they were non-porous and lacked the crystalline order of modern frameworks.



1990: The Conceptual Breakthrough

Hoskins & Robson

Proposed the revolutionary design of **3D scaffold-like frameworks**. This theoretical shift moved the field from simple polymers to engineered architectures.

Infinite Lattices

They envisioned a material where metal atoms act as "joints" and organic ligands act as "struts," creating an infinite, open molecular lattice.

1995: The Birth of MOF Synthesis + 1999: MOF-5 – The Landmark

Nobel Prize in Chemistry 2025



Omar Yaghi introduces hydrothermal methods to reliably engineer metal-organic bonds.

In 1999, Omar Yaghi and colleagues published the synthesis of **MOF-5**. This structure proved that high porosity and structural integrity could coexist without the framework collapsing upon solvent removal. MOF-5 effectively launched the modern era of porous materials research.

MOF-5 effectively launched the modern era of porous materials research.

